Complete Bounded Repair Ports:
Transfer, Reliability, and Geometric Structure

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Abstract

For a linear code and a distinguished coordinate, the complete bounded repair port records every dual-support recovery using at most a prescribed number of helpers. We keep three layers of this object visible: its support clutter, its scalar recovery coefficients, and its reliability under helper failures.

We prove an exact weighted-functional transfer theorem for concatenation and identify the persistent pointed obstruction to replicating a prescribed port. Every fixed represented port satisfying this obstruction bound occurs with positive density in an asymptotically good fixed-alphabet family. The same object has an exact deletion–contraction reliability calculus and a radius-truncated BEC EXIT hierarchy whose successive differences give the cheapest available repair radius. At full radius, repair reliability is a specialization of the Las Vergnas polynomial of the elementary perspective $M\backslash x \to M/x$; the bounded-radius filtration is extra information.

Two characteristic-three families illustrate the distinction. A twisted-cubic–axis code has exact coordinatewise matching and transversal rows and transfers strictly beyond the ordinary support-distance gate. A quartic normal rational curve with its nucleus has harmonic-quadruple circuits, a Steiner $S(3, 4, q + 1)$ nucleus port, a sparse Poisson repair window, and a series bottleneck at curve targets. Exact field-nine profiles make the support, reliability, EXIT, and pointed-Tutte boundaries explicit.

1 Introduction

Locality asks how many surviving symbols suffice to recover one unavailable coordinate. Availability asks for many disjoint choices of helpers. Both are standard in locally repairable codes [8, 14, 10], but neither describes the whole local failure event: two codes can have the same locality and matching number while their permitted repair sets overlap very differently.

The natural pointed object is the family of *all* bounded dual-support repairs. We call it a complete bounded repair port. Its matching number is disjoint availability, its transversal number minus one is worst-case helper failure tolerance, and its monotone Boolean function gives exact stochastic reliability. The coefficient layer matters as well: a repair support comes from a scalar equation, and concatenation is controlled by the minimum cost of realizing the corresponding block functional rather than by support alone.

The paper has six parts. Section 2 fixes the complete pointed object and separates support, coefficient, and probability layers. Section 3 gives exact and usable concatenation gates. Section 4 turns the pointed gate into a prescribed positive-density realization theorem. Section 5 develops reliability and bounded EXIT. Section 6 identifies the full-port polynomial as standard pointed-Tutte structure and isolates what the radius filtration adds. Section 7 compares cubic–axis and quartic–nucleus flagships.

The asymptotic coding ingredients, reliability identities, Chen–Stein approximation, and pointed-Tutte polynomial are classical. We develop their exact organization around the complete bounded pointed object, together with the transfer boundary, prescribed-port consequence, and the two exact geometric port inventories. We make no minimum-bandwidth claim, no full-MAP claim for a radius cutoff, and no threshold claim for harmonic closure. In the literature reviewed, we did not locate the exact complete-port confinement criterion or the two coordinatewise port inventories [14, 11, 9, 6]; this is a bounded comparison, not a priority claim.

2 Complete bounded repair ports

Let $C \leq \mathbb{F}_q^X$ be a linear code, let $x \in X$, and let $r \geq 0$.

Definition 2.1 (Support layer)**.** The complete radius-r repair hypergraph at x is

$$\mathcal{H}_x^{(\leq r)}(C) = \Big\{ R \subseteq X \setminus \{x\} : |R| \leq r,\ \exists w \in C^{\perp},\ w_x \neq 0,\ \operatorname{supp}(w) \setminus \{x\} = R \Big\}.$$

Its minimal members form the repair clutter $\min \mathcal{H}_x^{(\leq r)}(C)$. Write $\nu_x^{(\leq r)}$ and $\tau_x^{(\leq r)}$ for its matching and transversal numbers.

The distinction between the complete hypergraph and its minimal clutter is useful: the former records every bounded witness, while every monotone support invariant below depends only on the latter.

Proposition 2.2 (Basic invariants)**.** *Assume the empty helper set is not a repair. Then*

$$\nu_x^{(\leq r)} \leq \tau_x^{(\leq r)}.$$

Both numbers are unchanged after deleting nonminimal repair sets, so either the complete hypergraph or its minimal clutter may be used for these optima.

Proof. A matching of m nonempty edges forces every transversal to select at least one point from each edge. Minimalization preserves the upward closure of the success event, hence preserves matching and transversal optima. □

Definition 2.3 (Coefficient layer)**.** For $R \in \mathcal{H}_x^{(\leq r)}(C)$, its coefficient fiber is the set of normalized coefficient vectors

$$\left(-\frac{w_y}{w_x}\right)_{y \in R} \qquad (w \in C^{\perp},\ \operatorname{supp}(w) = R \cup \{x\}).$$

Each vector gives the exact recovery equation $c_x = \sum_{y \in R} (-w_y/w_x) c_y$. The direct protocol reads one complete $\mathbb{F}_q$ symbol from every helper; no subpacketized access or bandwidth optimum is asserted.

Definition 2.4 (Probability layer)**.** Let $A \subseteq X \setminus \{x\}$ be the surviving helpers and put

$$f_x^{(\leq r)}(A) = 1 \quad \iff \quad \exists R \in \mathcal{H}_x^{(\leq r)}(C) \text{ with } R \subseteq A.$$

For independent survival probabilities (s_y), the reliability is $R_x^{(\leq r)}(s) = \mathbb{E}[f_x^{(\leq r)}]$.

Exact equality of complete ports preserves all support-derived invariants: matching, transversals, blocker clutters, multivariate reliability, and every correlated-law reliability after transporting the helper labels. Coefficient fibers require the finer functional data used next.

3 Exact weighted-functional transfer

Let $I \le \mathbb{F}_q^m$ be an inner $[m,k,d]_q$ code and identify its message space with $L = \mathbb{F}_{q^k}$ through an $\mathbb{F}_q$-linear encoder $e : L \to I$. For $w \in \mathbb{F}_q^m$, define the induced block functional

$$\Phi_I(w) : L \longrightarrow \mathbb{F}_q, \qquad a \longmapsto \langle w, e(a) \rangle.$$

The zero fiber is $I^\perp$. For $\beta \in \mathrm{Hom}_{\mathbb{F}_q}(L, \mathbb{F}_q)$ put

$$\lambda(\beta) = \min_{\Phi_I(w)=\beta} \mathrm{wt}(w), \qquad \mu_x(\beta) = \min_{\substack{\Phi_I(w)=\beta \\ w_x \neq 0}} \mathrm{wt}(w),$$

with value ∞ for an empty constrained fiber.

Let $O \le L^J$ be an outer code. Its functional dual consists of tuples $(\beta_j)_{j\in J}$ satisfying

$$\sum_{j\in J} \beta_j(a_j) = 0 \qquad \text{for every } a = (a_j) \in O.$$

The weighted cost of such a tuple is $\sum_j \lambda(\beta_j)$. This is the ordinary quotient/coset-leader weight attached to the inner encoder; the pointed use below is the repair-specific part.

The all-zero functional sector has the fixed pointed cost

$$z_x(I) = \begin{cases} \mu_x(0) + d(I^\perp), & I^\perp \neq 0, \\ \infty, & I^\perp = 0. \end{cases}$$

Theorem 3.1 (Exact pointed confinement and transfer)**.** *Assume $|J| \ge 2$, and fix an outer block j and an inner coordinate x. The minimum weight of a concatenated dual witness through (j,x) that is not an embedded inner-dual witness is*

$$\min \left\{ z_x(I), \min_{\substack{(\beta_\ell) \text{ in the functional dual of } O \\ (\beta_\ell) \neq 0}} \left(\mu_x(\beta_j) + \sum_{\ell \neq j} \lambda(\beta_\ell) \right) \right\},$$

where a minimum over an empty set is ∞. Consequently, if this pointed cost is greater than $r+1$, then

$$\mathcal{H}^{(\le r)}_{(j,x)}(O \circ I) = \mathrm{embed}_j\big(\mathcal{H}^{(\le r)}_x(I)\big).$$

In particular it suffices that

$$r + 1 < 2d(I^\perp) \quad \textit{and} \quad \sum_j \lambda(\beta_j) \ge r + 2$$

for every nonzero functional-dual tuple.

Proof. A block tuple $w = (w_j)$ lies in $(O \circ I)^\perp$ exactly when $(\Phi_I(w_j))$ lies in the functional dual of O. Minimizing independently inside each non-target fiber gives λ; the target constraint gives μ_x. For a nonzero functional tuple this gives the inner minimum in the display. In the all-zero sector, a nonembedded witness needs a pointed inner-dual word in the target block and a nonzero inner-dual word in another block. Since $|J| \ge 2$, independent minimization gives the exact cost $z_x(I)$. This cost is at least $2d(I^\perp)$ whenever it is finite, giving the displayed sufficient bound. Below the minimum nonembedded weight every bounded pointed witness is confined to the target block, where it is exactly an embedded inner repair witness. $\square$

The functional-dual decomposition is classical concatenation machinery [3, Theorem 2.3][4, Theorem 2.1]. The weighted statement matters because ordinary functional support distance can be strictly weaker. For the completed field-nine cubic–axis inner code, a Singer-shifted generalized single-parity-check $[5,4,2]_{6561}$ outer code has functional distance five but weighted distance at least six. Its exact nonembedded threshold is six, so radius-four transfer holds although the support-distance gate requiring functional distance six fails. Here Singer-shifted means that the five parity-check multipliers are chosen through a regular Singer action so that the inner code's twenty unit-cost projective functional classes have disjoint translates in the five blocks. This finite calculation belongs to the checked transfer chain described in Section 8; Singer regularity is its only classical group input [16].

4 Prescribed positive-density realization

The pointed zero-functional sector persists even when outer dual distance grows.

Theorem 4.1 (Prescribed represented ports)**.** *Let $O_N \le L^N$ be L-linear outer codes with $d(O_N^{\perp}) \to \infty$, and let $C_N = O_N \circ I$. For all sufficiently large N, every pointed dual witness through (j,x) of weight at most $r+1$ is confined to block j if and only if*

$$r+1 < z_x(I).$$

Under this condition the radius-r port at x occurs at the N targets (j,x), hence with density $1/m$ in C_N.

Proof. The trace pairing identifies every $\mathbb{F}_q$-linear functional on L with $a \mapsto \mathrm{Tr}_{L/\mathbb{F}_q}(ba)$, and the functional-dual support distance equals the ordinary L-dual distance. Therefore every nonzero functional sector eventually costs more than $r+1$. The exact pointed decomposition in Theorem 3.1 leaves precisely the fixed zero-sector cost $z_x(I)$. Each outer block contains one copy of the target coordinate. □

Corollary 4.2 (Positive-density Clebsch repair ports)**.** *Let I be the $[6,3,4]_{11}$ Clebsch hexagon code [15], and fix a coordinate x. The minimal radius-three repair clutter at x is the complete three-uniform hypergraph on the other five coordinates. Hence it has ten minimal repair supports,*

$$\nu_x^{(\le 3)} = 1, \qquad \tau_x^{(\le 3)} = 3,$$

and homogeneous helper-survival reliability

$$10s^3(1-s)^2 + 5s^4(1-s) + s^5.$$

The full radius-five coefficient port at x determines I. Moreover $z_x(I) = 8$, so this entire coefficient-valued port occurs with density $1/6$ in an asymptotically good fixed-$\mathbb{F}_{11}$ family.

Proof. The dual $I^{\perp}$ is again an $[6,3,4]_{11}$ MDS code. Every four-set containing x is therefore the support of a dual word, unique up to scalar. The minimal helper sets are consequently all $\binom{5}{3} = 10$ triples on the other coordinates. Their matching and transversal numbers are one and three, and recovery succeeds exactly when at least three of the five helpers survive, giving the displayed polynomial.

At radius five the coefficient fibers contain every normalized dual word with nonzero x-coordinate. These vectors span $I^{\perp}$: after normalizing the x-coordinate to one, their differences span the kernel of evaluation at x, and any one normalized vector supplies the remaining direction. Thus the coefficient port recovers $I^\perp$ and hence I.

Finally, the minimum dual-word weight through x is four, so

$$z_x(I) = \mu_x(0) + d(I^\perp) = 4 + 4 = 8.$$

Taking $r = 5$ gives $r + 1 = 6 < 8$. Theorem 4.1 therefore confines every full pointed witness to its inner block and reproduces the coefficient fibers at one of the six coordinate types in every block. Choosing asymptotically good outer codes over $\mathbb{F}_{11^3}$ gives the claimed fixed-$\mathbb{F}_{11}$ family and density. $\square$

Remark 4.3. The support clutter and reliability polynomial in Corollary 4.2 are shared by every $[6, 3, 4]$ MDS code. The Clebsch-specific information lies in the scalar coefficient fibers: they recover the inner code that is characterized geometrically in [15].

This is a represented-port theorem: it does not say that every abstract hypergraph is linearly representable. It yields standard scaled asymptotic regions. If $Q = q^k$, random L-linear outer codes of rate R can have relative distances $\delta, \delta^\perp$ whenever

$$H_Q(\delta) < 1 - R, \qquad H_Q(\delta^\perp) < R.$$

For an inner $[m, k, d]_q$ code the concatenated parameters satisfy

$$R_{\text{concat}} \to \frac{k}{m} R, \qquad \delta_{\text{concat}} \geq \frac{d}{m}\delta.$$

Indeed, for a random L-linear outer code the expected numbers of nonzero words and dual words below the two cutoffs have exponential rates $R + H_Q(\delta) - 1$ and $H_Q(\delta^\perp) - R$, respectively, so both vanish simultaneously [12]. When Q is square, algebraic-geometry outer codes similarly give, for $\gamma = 1/A(Q)$ and $\gamma < R < 1 - \gamma$,

$$\delta(O_N) \geq 1 - R - \gamma, \qquad \delta(O_N^\perp) \geq R - \gamma, \qquad \delta_{\text{concat}} \geq \frac{d}{m}(1 - R - \gamma).$$

These are the classical GV and TVZ ingredients. For the completed $[20, 4, 9]_9$ seed, $Q = 6561$ and $\gamma \leq 1/80$, recovering the concrete rate-$1/10$ family as one instance [18, 17].

5 Reliability and bounded EXIT

Let $\mathcal{H}$ be a repair clutter on helper set V. For $v \in V$, define

$$\mathcal{H} - v = \min\{E \in \mathcal{H} : v \notin E\}, \qquad \mathcal{H}/v = \min\{E \setminus \{v\} : E \in \mathcal{H}\}.$$

Theorem 5.1 (Reliability calculus)**.** *For independent helper survival probabilities (s_v),*

$$R_\mathcal{H}(s) = (1 - s_v) R_{\mathcal{H}-v}(s) + s_v R_{\mathcal{H}/v}(s),$$

and

$$\frac{\partial R_\mathcal{H}}{\partial s_v} = R_{\mathcal{H}/v} - R_{\mathcal{H}-v} = \Pr(v \textit{ is pivotal}).$$

Under homogeneous survival s, $R'_\mathcal{H}(s) = \sum_v \Pr_s(v$ is pivotal$)$. If the minimum blocker size is τ and there are b_τ minimum blockers, then for helper failure probability p,

$$1 - R_\mathcal{H}(1 - p) = b_\tau p^\tau + O(p^{\tau+1}).$$

Proof. Condition on whether v survives. Differentiation gives pivotality and the homogeneous identity. Failure occurs exactly when the failed helper set contains a minimal blocker; inclusion–exclusion begins with the minimum blockers and all intersections contribute only at higher order. $\square$

These are the standard deletion–contraction and pivotality identities for a monotone reliability system [5].

For EXIT conventions, condition the target x unavailable and erase helper v with probability p_v. Put

$$h_x^{(\le r)}(p) = \Pr(f_x^{(\le r)}(A) = 0), \qquad R_x^{(\le r)} = 1 - h_x^{(\le r)}.$$

This is an *extrinsic* failure probability. If the target is itself erased with probability p_x, its residual erasure probability is $p_x h_x^{(\le r)}$. At full radius it is symbol-MAP EXIT; at finite radius it is a bounded-query decoder and carries no capacity claim [2, 13].

Proposition 5.2 (Bounded-EXIT hierarchy)**.** *The failure form of deletion–contraction is*

$$h_{\mathcal{H}}(p) = p_v h_{\mathcal{H}-v}(p) + (1-p_v) h_{\mathcal{H}/v}(p), \qquad \frac{\partial h_{\mathcal{H}}}{\partial p_v} = \Pr(v \textit{ is pivotal}).$$

If $L_x(A)$ is the size of the cheapest repair contained in the realized survivor set, with $L_x = \infty$ when none exists, then

$$\Pr(L_x = r) = h_x^{(\le r-1)} - h_x^{(\le r)}, \qquad \Pr(L_x = \infty) = h_x^{\mathrm{MAP}}.$$

Minimal transversals of $\mathcal{H}_x^{(\le r)}$ are exact target-specific, one-shot bounded-repair failure certificates. They are not automatically the representation-dependent stopping sets of a global iterative Tanner decoder.

For a length-N code define the radius-r locality deficit

$$L_r(x) = \int_0^1 \big(h_x^{(\le r)}(p) - h_x^{\mathrm{MAP}}(p)\big)\, dp.$$

If Δa_k counts size-k survivor sets repaired by full MAP but not at radius r, beta integration gives

$$L_r(x) = \sum_{k=0}^{N-1} \frac{\Delta a_k}{N\binom{N-1}{k}}.$$

With entropy normalized in $\log_q$ units, the unrecoverable-symbol indicator is the normalized conditional entropy on the q-ary erasure channel. Under this convention the symbol-MAP areas sum to the code dimension K [2], so

$$\sum_x \int_0^1 h_x^{(\le r)}(p)\, dp = K + \sum_x L_r(x).$$

6 Pointed Tutte structure

Let M be the column matroid of a projective-system code on $E = V \sqcup \{x\}$, with x neither a loop nor a coloop. Put

$$\epsilon_x(A) = r_M(A \cup \{x\}) - r_M(A) \in \{0, 1\}.$$

The set A repairs x exactly when $\epsilon_x(A) = 0$.

Theorem 6.1 (Pointed perspective polynomial)**.** *Let $D = M\backslash x$ and $Q = M/x$. The Las Vergnas polynomial of the elementary perspective $D \to Q$ is*

$$T_{D\to Q}(X,Y,Z) = \sum_{A\subseteq V} (X-1)^{r(M)-r_M(A\cup\{x\})}(Y-1)^{|A|-r_M(A)}Z^{\epsilon_x(A)}.$$

If

$$S_x(u) = \sum_{\epsilon_x(A)=0} u^{|A|},$$

then, for $n = |V|$,

$$R_{(M,x)}(s) = (1-s)^n S_x\left(\frac{s}{1-s}\right).$$

Writing $U_K(u,y) = \sum_A u^{|A|}y^{r_K(A)}$, one also has

$$S_x(u) = \frac{\partial}{\partial y}\big(U_D(u,y) - U_Q(u,y)\big)\Big|_{y=1}.$$

Proof. Contraction gives $r_D(A) = r_M(A)$ and $r_Q(A) = r_M(A\cup\{x\}) - 1$, so $r_D(A) - r_Q(A) = 1 - \epsilon_x(A)$. Substitution in the rank-one perspective expansion gives the displayed polynomial. The Z^0 terms are exactly the successful sets; more explicitly,

$$S_x(u) = u^{r(M)}[Z^0]T_{D\to Q}(1+u^{-1}, 1+u, Z),$$

which yields the reliability formula. Finally,

$$\partial_y(y^{r_D(A)} - y^{r_Q(A)})\Big|_{y=1} = r_D(A) - r_Q(A),$$

which is the zero–one success indicator. □

This identification is the singleton specialization of standard set-pointed Tutte theory [20, equation (3.1)]; it is not a new polynomial. Pointed duality gives

$$R_{(M,x)}(s) + R_{(M^*,x)}(1-s) = 1,$$

and the minimal failure blockers of (M,x) are exactly the minimal repair sets of (M^*,x). For a represented matroid they are the sets $Q' \setminus \{x\}$, where Q' ranges over cocircuits containing x; equivalently, Q' is an inclusion-minimal row-code support through x. In particular,

$$\tau_{\text{full}}(x) = d_x(C) - 1, \qquad d_x(C) = \min\{\text{wt}(c) : c \in C,\ c_x \neq 0\}.$$

The full pointed polynomial forgets the repair radius. Six-element circuits in a rank-five code create five-helper repairs that are absent from a radius-four port. Thus the hierarchy in Section 5 is a genuine filtered refinement of the standard perspective invariant.

7 Geometric flagships

7.1 The twisted-cubic–axis family

Let $q = 3^h$ and work in $\mathrm{PG}(3, q)$. Put

$$T_q = \{C(t) = (1 : t : t^2 : t^3) : t \in \mathbb{F}_q\},$$

$$L_q = \{A(u) = (0 : 1 : u : 0) : u \in \mathbb{F}_q\} \cup \{A_\infty = (0 : 0 : 1 : 0)\}.$$

The line L_q is the characteristic-three axis of the osculating-plane pencil [6]; all statements below follow directly from the displayed coordinates.

Theorem 7.1 (Cubic–axis port)**.** *The row code on $T_q \sqcup L_q$ is $[2q+1, 4, q-1]_q$. Its minimal circuits of size at most four are the triples on L_q and the sets*

$$\{C(s), C(t), C(u), (0 : e_1 : e_2 : 0)\}$$

for distinct s, t, u, where $e_1 = s + t + u$ and $e_2 = st + su + tu$. Every cubic coordinate has

$$(\nu, \tau) = \left(\frac{q-1}{2}, q-2\right).$$

If $Z_3(q)$ is the maximum size of a three-term-zero-sum-free subset of $(\mathbb{F}_q, +)$, then

$$(\nu, \tau)_{A_\infty} = \left(\frac{5q-3}{6}, 2q - 1 - Z_3(q)\right),$$

$$(\nu, \tau)_{A(a)} = \left(\frac{5q-9}{6}, 2q - 2 - Z_3(q)\right).$$

In particular every coordinate satisfies $\tau > \nu$ for $q \geq 9$.

Proof sketch. A plane containing L_q contains at most one cubic point by Frobenius injectivity; a plane not containing it meets the line once and the cubic at most three times. This gives the code parameters. Vandermonde determinants classify the displayed small circuits. At an axis target the repair clutter is a disjoint union of K_q and a zero-sum-triple hypergraph, so matching and transversal numbers add; complements of transversals are precisely zero-sum-free sets. At a cubic target, inversion relabels the other cubic helpers by $\mathbb{F}_q^*$. A multiplicative-generator pairing gives $(q-1)/2$ repairs with distinct axis completions; no larger matching is possible because each repair uses two of the $q-1$ cubic helpers. The shifted-inverse complement lemma bounds a repair-free helper set by $q+2$, attained by all axis helpers together with one cubic helper. Taking complements gives $\tau = q-2$. The stated strict inequalities follow from the elementary bound $Z_3(q) \leq q-2$ and the displayed formulas. □

At $q = 9$, $Z_3(9) = 4$, and the three coordinate types have exact rows

$$(4, 7), \qquad (6, 12), \qquad (7, 13),$$

with multiplicities $9, 9, 1$. Restoring the cubic point at infinity gives a $[2q+2, 4, q]_q$ code whose full minimal port is visible at radius four and has uniform rows

$$\left(\frac{q-1}{2}, q-1\right) \text{ on the cubic,} \qquad \left(\frac{5q-3}{6}, 2q-3\right) \text{ on the axis.}$$

This is the inner code in the strict weighted-transfer example above.

7.2 The quartic–nucleus harmonic family

Let

$$V(t) = (1, t, t^2, t^3, t^4), \qquad V(\infty) = e_4, \qquad N = e_2,$$

and let $S_q = \{V(t) : t \in \mathbb{P}^1(\mathbb{F}_q)\} \cup \{N\}$. The classical normal-rational-curve nucleus formula identifies N as the hyperplane nucleus in characteristic three [7].

Theorem 7.2 (Quartic–nucleus port)**.** *For $q = 3^h \geq 9$, the row code Q_q on S_q has parameters*

$$[q + 2, 5, q - 3]_q, \qquad d(Q_q^{\perp}) = 5.$$

Its minimal circuits of size at most five are exactly

$$\{N\} \cup B,$$

where B is a harmonic quadruple of $\mathbb{P}^1(\mathbb{F}_q)$. These quadruples form a Steiner system $S(3, 4, q+1)$, the characteristic-three four-set orbit in the standard $\mathrm{PGL}(2, q)$ design construction [19, Section 2]. Hence the radius-four repairs are all B at the nucleus and all $\{N\} \cup (B \setminus \{x\})$ at a curve target $x \in B$. Every curve target has $(\nu, \tau) = (1, 1)$. At $q = 9$ the nucleus has $(\nu, \tau) = (2, 5)$, with 30 repairs, while each curve target has 12.

Proof. For distinct finite a, b, c, d, the hyperplane through their quartic columns contains N exactly when $e_2(a, b, c, d) = 0$; with ∞ present the condition is $a + b + c = 0$. Given three finite points, if $e_1 = a + b + c \neq 0$, the unique completion is $d = -e_2/e_1$; if $e_1 = 0$, the unique completion is ∞. The completion is distinct from the original triple. This proves the Steiner property and identifies the blocks as the projective images of $\{\infty, 0, 1, -1\}$.

Any five curve points are independent. The determinant criterion shows that $\{N\} \cup B$ has rank four exactly for a harmonic block, and deleting any point restores independence. Thus there are no smaller circuits and the dual distance is five. A hyperplane contains at most four curve points and possibly N, with equality on a harmonic block, so the primal distance is $(q + 2) - 5 = q - 3$. The repair and row statements follow from the circuit inventory; the field-nine nucleus row is an exact finite enumeration. $\square$

The inner-dual half of the coarse transfer gate is automatic at radius four because $5 < 2d(Q_q^{\perp}) = 10$. Theorem 4.1 therefore places the entire harmonic port with positive density in asymptotically good fixed-$\mathbb{F}_q$ families.

7.3 Reliability, EXIT, and the geometric contrast

At $q = 9$, write the homogeneous reliability in Bernstein form

$$R(s) = \sum_{k=0}^{10} a_k s^k (1 - s)^{10-k}.$$

The harmonic nucleus and curve profiles are

$$(a_0, \ldots, a_{10})_N = (0, 0, 0, 0, 30, 180, 210, 120, 45, 10, 1),$$

$$(a_0, \ldots, a_{10})_{\mathrm{curve}} = (0, 0, 0, 0, 12, 72, 126, 84, 36, 9, 1).$$

The nucleus has 72 minimum blockers of size five; the nucleus singleton is the unique minimum blocker at every curve target. The exact radius-four EXIT deficits are

$$L_4(N) = \frac{2}{77}, \qquad L_4(\text{curve}) = \frac{23}{154},$$

and the total radius-four area is

$$5 + \frac{2}{77} + 10\frac{23}{154} = \frac{502}{77}.$$

The full pointed-Tutte success enumerators add repairs coming from larger circuits:

$$S_N(u) = 30u^4 + 252u^5 + 210u^6 + 120u^7 + 45u^8 + 10u^9 + u^{10},$$

$$S_{\text{curve}}(u) = 12u^4 + 234u^5 + 210u^6 + 120u^7 + 45u^8 + 10u^9 + u^{10}.$$

Their difference from the truncated profiles is the concrete radius-filtration boundary promised in Section 6.

There is also an asymptotic reliability contrast. Along any sequence of Steiner systems $S(3,4,n)$ with $n \to \infty$, retain each point independently with probability $s = cn^{-3/4}$. If X counts retained blocks, dependency-graph Chen–Stein approximation gives

$$X \implies \text{Poisson}(c^4/24), \qquad \Pr(X > 0) \longrightarrow 1 - e^{-c^4/24},$$

with total-variation error $O(n^{-1/4})$ [1]. This is the repair window at the nucleus. Indeed, there are $b = n(n-1)(n-2)/24$ blocks; each block meets $3(n-4)$ others in two points and $2(n-4)(n-8)/3$ others in one point. The dependency bound has joint terms $O(n^4 s^6)$ and $O(n^5 s^7)$, giving $O(n^{-1/4})$ at the displayed scale. At a curve target the blocks through it induce an $S(2,3,n-1)$, but every repair also uses N. If N survives with probability ρ and curve helpers survive with probability $c(n-1)^{-2/3}$, repairability tends to

$$\rho\big(1 - e^{-c^3/6}\big),$$

with conditional Poisson error $O(n^{-1/3})$. Thus the nucleus target has a sparse parallel window, while every curve target has a series bottleneck. For $m = n-1$, the derived design has $m(m-1)/6$ triples and $m(m-1)(m-3)/8$ unordered pairs meeting in one point, which gives the stated conditional error rate from the same bound.

Finally, the harmonic small-circuit closure has a deterministic nucleus gate. Let $D(S)$ close curve set S under "three points of a harmonic block add the fourth." Then

$$\text{cl}_4(S) = S \quad \text{if } S \text{ contains no block}, \qquad \text{cl}_4(S) = \{N\} \cup D(S) \quad \text{otherwise},$$

and $\text{cl}_4(S \cup \{N\}) = \{N\} \cup D(S)$. At $q = 9$ there is a block-free rank-five set of five curve points that linearly spans all eleven points but is inert under cl_4; adjoining N completes all five missing curve points in one parallel round. This is a finite deterministic separation, not an asymptotic propagation or threshold theorem.

8 Verification and provenance

The finite transfer core and the cubic–axis theorem chain are checked in Lean. The paper-facing roots are `RepairCodes` and `RepairPorts.FunctionalCost`; the exact public module closure and artifact provenance are release-gate data, not claims made by this private draft. The asymptotic trace, random-code, and AG arguments remain manuscript-level arguments.

The harmonic and probability statements have deterministic independent replays. The load-bearing private evidence bundle consists of the scripts and JSON certificates for C218 (quartic circuits), C219 (reliability), C226 (bounded EXIT transforms), C227 (pointed-Tutte profiles), C243 (nucleus gate), and C244 (pointed distances, EXIT deficits, and Poisson rates). Each report records its command, conventions, checked counts, hashes, and finite boundary. The field-nine checks do not imply unproved all-field classifications; the all-field statements above have symbolic proofs.

The only nonformalized axiom in the cited Lean theorem chain is Stichtenoth's self-dual TVZ-family theorem over $\mathbb{F}_{6561}$ [17]. Classical inputs are labeled as such: concatenated-dual decomposition, Singer regularity, normal-rational- curve nuclei, reliability/EXIT calculus, Chen–Stein approximation, and the Las Vergnas perspective polynomial.

9 Conclusion

A complete bounded repair port is one pointed object with three compatible views. Its support clutter measures disjoint availability and adversarial tolerance; its coefficient fibers give exact scalar recovery and the correct weighted transfer cost; its Boolean function gives reliability and bounded EXIT. Exact concatenation transfers the whole object, and the persistent pointed obstruction precisely controls positive-density realization.

The cubic–axis family shows rich overlapping ports and strict weighted transfer beyond a support-only gate. The quartic–nucleus family replaces that overlap pattern by a harmonic Steiner design: the nucleus has a sparse parallel repair window while curve targets share a compulsory series helper. Pointed-Tutte theory explains the full-radius structure, while the explicit field-nine profiles isolate what the bounded-radius truncation retains and what the full-radius invariant adds.

A sharper cubic blocker-stability statement is not needed for the present results. Sequential composition, general service regions, coefficient optimization, log-concavity, and random harmonic-cascade thresholds lie outside this paper.

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